Aadhav S. Bharadwaj

Cleveland, OH (Open to Relocation) | 804-836-0862 | bharadwajaadhav@gmail.com
US Citizen | GitHub | LinkedIn | Portfolio

Education

Case Western Reserve University Aug 2023 – May 2027
B.S. Computer Science; Secondary Major in Mathematics; Minor in Economics *Cleveland, OH*
GPA: 3.9/4.0 • Dean's High Honors List • Merit Scholarship

Relevant coursework: Machine Learning, Artificial Intelligence, Algorithms, Database Systems, Operating Systems, Efficient Deep Learning, Computer Security, Linear Algebra, Statistics

Minor in Economics: Microeconomics, Macroeconomics

London School of Economics and Political Science (LSE) Summer 2025
Study Abroad — Economics *London, UK*

Technical Skills

Languages: Python, TypeScript/JavaScript, Java, SQL, R
AI/ML: RAG, Embeddings, LLM-as-Judge Evals, CI Eval Gates, Hybrid Retrieval (BM25 + RRF), Elasticsearch, pgvector, PyTorch
LLM Infrastructure: Langfuse, vLLM, AWS Bedrock, FastAPI, OpenAI-Compatible Gateways
Full Stack: Next.js, React, Node.js, Express, MongoDB, PostgreSQL
Real-Time / Infra: Apache Spark (PySpark), Socket.IO, WebRTC, mediasoup, Docker, CI/CD, AWS (Lambda, S3), Linux

Experience

Abeyon May 2026 – Present
Software Engineer Intern – AI/LLM *Washington, DC*
- Deployed self-hosted Langfuse v3 and instrumented a production FastAPI LLM gateway (vLLM + AWS Bedrock) with tracing, custom pricing across 9 models, and structured logging for company-wide cost, token, and latency attribution.
- Built LLM-as-judge evaluation pipelines with frozen gold sets; executed 100-case benchmarks and stage-level RAG debugging.
- Implemented BM25 + dense hybrid retrieval with Reciprocal Rank Fusion in Elasticsearch, improving Recall@40 from 0.875 to 0.887 on labeled evaluation sets.
- Reduced work-item generation latency from 123.6s to 49.7s (2.49×) via parallel NDJSON batch processing and migrated inference routing to AWS Bedrock across three applications.

Eaton Corporation Sep 2025 – Jan 2026
AI Software Engineer Intern (contracted through xLab) *Remote*
- Architected GenAI RAG chatbots for customer-interaction training using a MERN backend with vector retrieval and embedding search, improving onboarding efficiency by approximately 40%.
- Deployed containerized MERN platform with CI/CD, JWT/RBAC, and live WebSocket KPI dashboards for production trainer session analytics across modular training workflows.

WSOM, Case Western Reserve University Jun 2025 – Present
Software Engineer Intern (AI Avatar Kiosk) *Cleveland, OH*
- Built a dual-screen AI avatar kiosk using Next.js, AWS Lambda, HeyGen, and Pinecone-backed RAG; achieved sub-1.5s latency while serving 100+ daily users through CWRU SSO.

CWRU Computer Science Department Aug 2024 – Present
Undergraduate Teaching Assistant *Cleveland, OH*
- Lead weekly CS labs for 100+ students, hold 5+ office hours weekly, and developed Python autograding utilities supporting course operations.

Wildlife SOS Jun 2025 – Jul 2025
Software Developer *Remote*
- Developed an internal asset-tracking platform supporting 1,000+ users across 12 conservation sites using Next.js, MongoDB, and Vercel.

Projects

Blackbox-QA | Python, FastAPI, pgvector, Langfuse, Docker
- Built a framework-free tool-calling agent (bounded ReAct loop, 3 tools, read-only SQL guard) over hybrid retrieval (BM25 + dense + RRF, cross-encoder rerank) across a 30,646-report / 125,213-chunk corpus; lifted Recall@5 from 0.04 (BM25-only) to 0.64 and MRR to 0.62.
- Instrumented self-hosted Langfuse tracing with LLM-as-judge scoring via the Scores API; measured citation_match of 0.68 and answer_quality of 0.76 on a frozen gold set.
- Gated a query-rewrite retry on a calibrated cross-encoder score (TPR 1.0 / FPR 0.05) and added a deterministic Recall@5 CI regression gate on every pull request.

Arguably | mediasoup, WebRTC, Socket.IO, Next.js
- Implemented a standalone mediasoup SFU and Socket.IO signaling stack (2.5k LOC) and integrated browser WebRTC functionality through custom media components.
- Added SFU unit testing, CI automation, HTTPS/CORS/TURN infrastructure, and production reliability improvements.

Palate | MERN, GPT-4
- Developed an AI recipe generation platform serving 500+ generated recipes with 98% uptime.

Awards

HackCWRU 2025 — 4th Place Overall, ExpressInk (AI Drawing Analysis)